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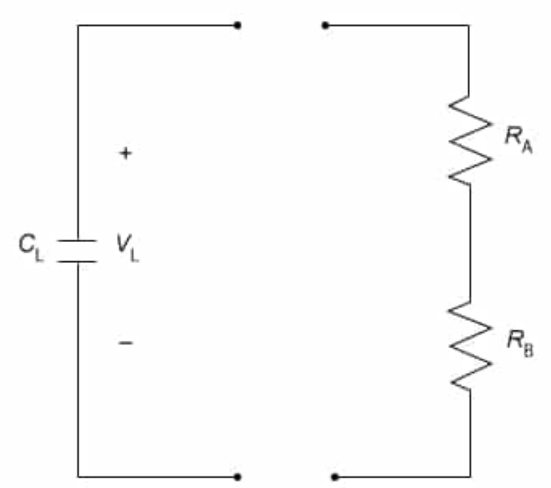

Figure 1 Circuit model of lightning passing through the air and the body

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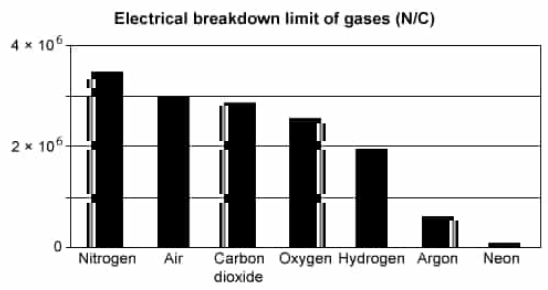


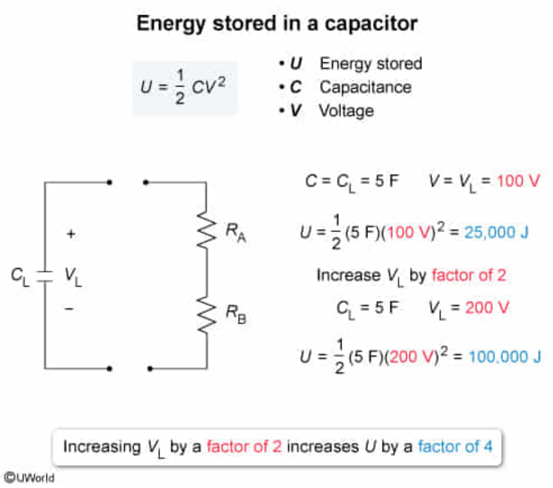
Figure 2 Breakdown limit of common gases (in N/C)

Which of the following changes will increase the energy stored in the capacitor in the lightning circuit model by the greatest amount?

- A. Increasing *C* by a factor of 2 [25%]
- B. Decreasing *C* by a factor of 2 [4%]
- ✔ C. Increasing *V* by a factor of 2 [59%]
- D. Decreasing *V* by a factor of 2 [11%]

Correct
59% answered correctly

Explanation:



The **energy** *U* stored in a **capacitor** is equal to half the product of the **capacitance** *C* and the square of the voltage *V*:

$$U = \frac{1}{2}CV^2$$

In this question, a 5 F capacitor is charged to 100 V, giving a *U* of:

$$U = \frac{1}{2}(5\text{ F})(100\text{ V})^2 = 25{,}000\text{ J}$$

Because *U* is directly proportional to *C*, increasing or decreasing *C* by a factor of 2 will increase or decrease *U* by that same factor. However, *U* is proportional to the square of *V*, so doubling *V* yields:

$$U = \frac{1}{2}(5\ \text{F})(200\ \text{V})^2 = 100{,}000\ \text{J}$$

Therefore, increasing *V* by a factor of 2 causes *U* to increase by a factor of 4.

(Choice A) Increasing the capacitance by a factor of 2 only increases the energy stored by a factor of 2.

(Choices B and D) Decreasing the capacitance or the voltage causes the energy stored to decrease, not increase.

Educational objective:
The potential energy stored in a capacitor equals half the product of the capacitance and the square of the voltage. Increasing the voltage causes a larger increase in the potential energy compared to a similar increase in the capacitance.

Time spent: 0 sec
QID: 403731
Subject: Physics
System: Electrostatics and Circuits

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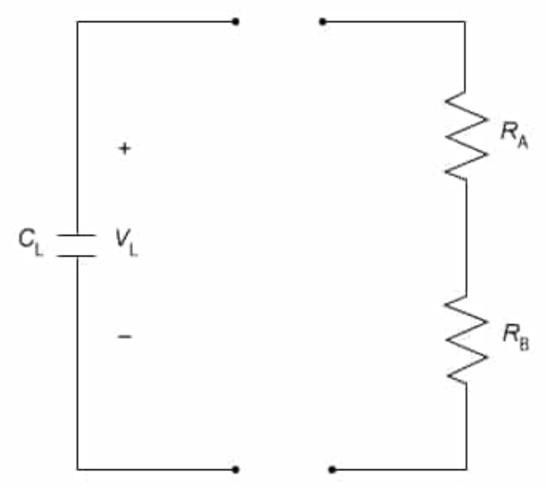

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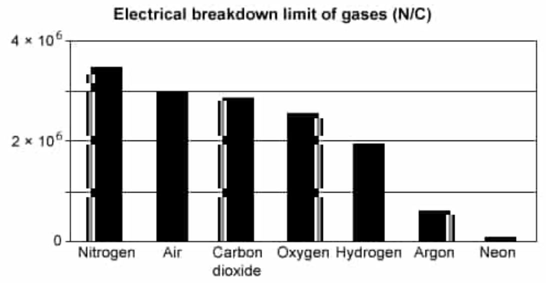


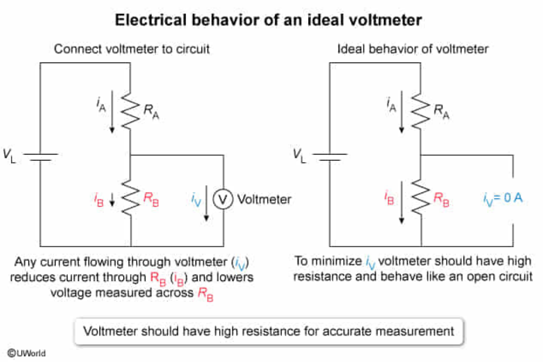
Figure 2 Breakdown limit of common gases (in N/C)

A voltmeter is used to measure the voltage across R_B in Figure 1 after the switch is closed. Which of the following best describes the electrical properties of the voltmeter?

- A. It has a very large resistance. [39%]
- B. It has a resistance similar to R_B. [15%]
- C. It draws a very large current. [11%]
- D. It draws a current similar to R_B. [33%]

Correct
38% answered correctly

Explanation:



A **voltmeter** measures the voltage V between two points in an electric circuit. To measure V across a **resistor**, the voltmeter is connected in parallel with the resistor because circuit elements connected in parallel have the same V. By Ohm's law, the resistor's V equals the product of the current I and the **resistance** R:

$$V = \mathrm{IR}$$

To achieve an accurate measurement of V, connecting the voltmeter to the circuit should not alter the current through the resistor.

In this question, the voltmeter measures V across R_B and should not affect the current I through R_B. However, a voltmeter with a low resistance will draw a large current, decreasing the I through R_B. Consequently, the voltmeter will affect the circuit and measure a lower V across R_B. Therefore, the voltmeter should have a very large resistance to minimize any changes to I through R_B. A voltmeter with a very large resistance behaves like an open circuit (ie, no current flows through the voltmeter) when it is connected to a circuit.

(Choice B) If the voltmeter's resistance is similar to R_B, some of the current that should flow through R_B instead flows through the voltmeter, and the voltage measured across R_B is lower than the true value.

(Choices C and D) If the voltmeter draws either a large current or a current similar to R_B, the current through R_B is reduced and the voltage measured across R_B is lower than the true value.

Educational objective:
A voltmeter is connected in parallel with a circuit element to measure its voltage. The voltmeter should behave like an open circuit (ie, have a very large resistance) to ensure accurate voltage measurements.

Time spent: 0 sec
Subject: Physics

QID: 403732
System: Electrostatics and Circuits

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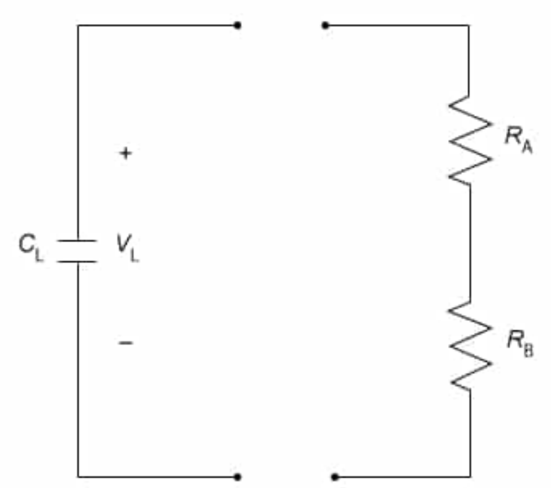

Figure 1 Circuit model of lightning passing through the air and the body

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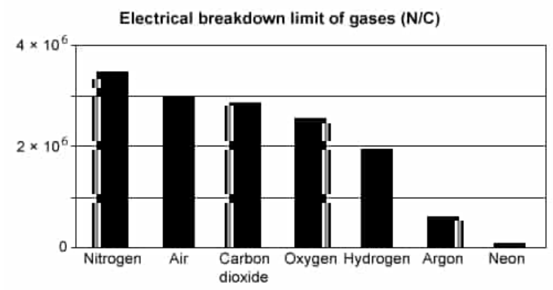


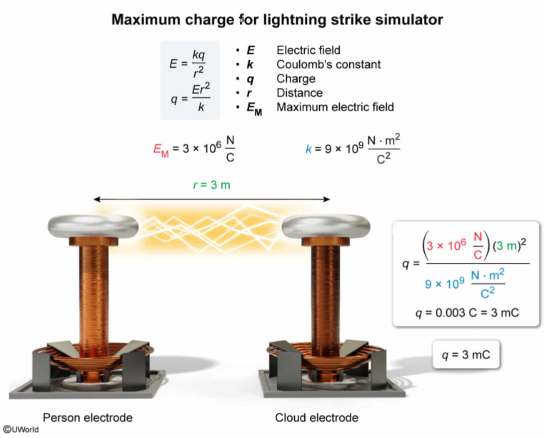
Figure 2 Breakdown limit of common gases (in N/C)

For the lightning strike simulator described in the passage, the distance between the person electrode and the cloud electrode is 3 m. Assume that the electrodes behave like point charges. What is the maximum charge on the cloud electrode when the electrical breakdown limit of air is reached? (Note: Coulomb's constant is $k = 9 \times 10^9$ N·m²/C².)

- A. 9 mC [8%]
- B. 3 mC [60%]
- C. 1 mC [20%]
- D. 0.3 mC [10%]

Correct
59% answered correctly

Explanation:



The electric field E produced by a **charge** q depends on Coulomb's constant k and the **distance** r from q, according to the equation:

$$E = \frac{kq}{r^2}$$

This equation can be rearranged to solve for q, yielding:

$$q = \frac{Er^2}{k}$$

Air is a good electrical insulator until E reaches its electrical breakdown limit E_M. From Figure

2, E_M for air is 3 × 10^6 N/C. When E_M is reached, air becomes a conductor of electricity and a lightning strike can occur.

In this question, the person electrode and cloud electrode of the lightning strike simulator are separated by 3 m, and E at the person electrode is E_M. Substituting the values into the equation for q gives:

$$q = \frac{\left(3 \times 10^6 \frac{\text{N}}{\text{C}}\right)(3\text{ m})^2}{\cancel{9} \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}} = \frac{\left(3 \times 10^6\right)(\cancel{9})}{\cancel{9} \times 10^9}\left(\frac{\cancel{\text{N} \cdot \text{m}^2}}{\cancel{\text{C}}}\right)\left(\frac{\text{C}^{\cancel{2}}}{\cancel{\text{N} \cdot \text{m}^2}}\right)$$

$$q = 3 \times 10^{(6-9)}\text{ C} = 0.003\text{ C} = 3\text{ mC}$$

Therefore, q equals 3 mC when E equals E_M and a lightning strike occurs.

(Choice A) 9 mC results from using the distance cubed in the electric field equation instead of the distance squared.

(Choice C) 1 mC is obtained by not squaring the distance term in the electric field equation.

(Choice D) 0.3 mC is calculated if the distance squared term is not included in the electric field equation.

Educational objective:

The electric field is calculated from the equation $E = \frac{\text{kq}}{r^2}$. When E exceeds the electrical breakdown limit of air, the air becomes an electrical conductor and lightning strikes can occur.

Time spent:0 sec
Subject:
Physics

QID:403733
System:
Electrostatics and Circuits

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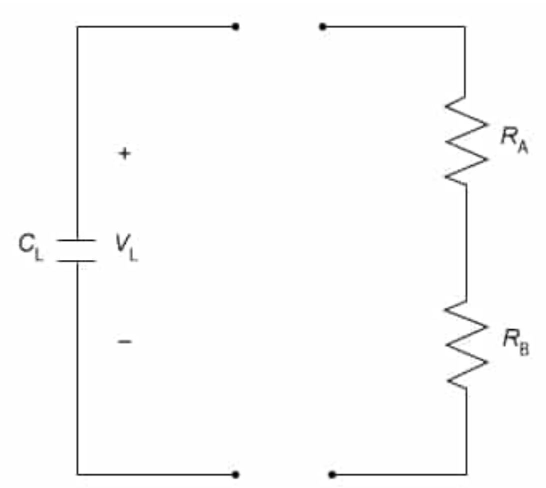

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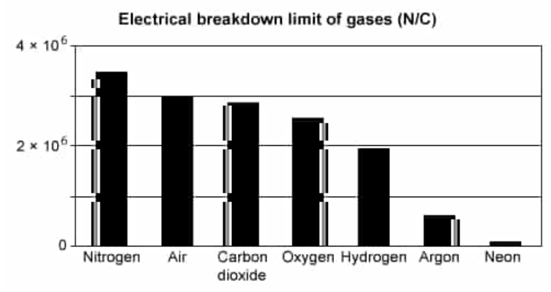


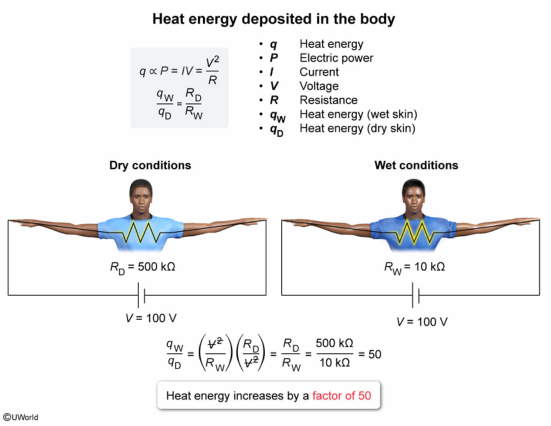
Figure 2 Breakdown limit of common gases (in N/C)

A voltage of 100 V is applied from hand to hand. How many times greater is the heat energy deposited in the body under wet conditions compared to dry conditions?

- A. 5 [5%]
- B. 10 [5%]
- C. 50 [85%]
- D. 100 [3%]

Correct
86% answered correctly

Explanation:



The **heat energy** *q* generated by a **resistor** is proportional to its electric power *P*:

$$q \propto P$$

P equals the product of the current *I* and the voltage *V* across the resistor:

$$P = \mathrm{IV}$$

By Ohm's law, *V* across a resistor equals the product of the current *I* and the **resistance** *R*:

$$V = \mathrm{IR}$$

Ohm's law can be rearranged to solve for *I*:

$$I = \frac{V}{R}$$

Combining this equation with the definition of *P* yields:

$$P = \left(\frac{V}{R}\right)V = \frac{V^2}{R}$$

Therefore, q is proportional to the ratio of V squared and R:

$$q \propto \frac{V^2}{R}$$

In this question, Table 1 lists R from hand to hand as 500 kΩ under dry conditions and 10 kΩ under wet conditions. Hence, q with dry skin q_D and 100 V applied from hand to hand is proportional to:

$$q_D \propto \frac{(100\text{ V})^2}{500\text{ k }\Omega}$$

Furthermore, q with wet skin q_W is proportional to:

$$q_W \propto \frac{(100\text{ V})^2}{10\text{ k }\Omega}$$

The ratio of q_W and q_D equals:

$$\frac{q_W}{q_D} = \left(\frac{\cancel{(100\text{ V})^2}}{\cancel{10\text{ k }\Omega}}\right)\left(\frac{50\cancel{0}\ \cancel{\text{k }\Omega}}{\cancel{(100\text{ V})^2}}\right) = 50$$

Therefore, q_W is 50 times larger than q_D.

(Choice A) A 5-times greater heat deposition is calculated by using 100 kΩ for the resistance from hand to hand under wet conditions instead of 10 kΩ.

(Choice B) A 10-times greater heat deposition is obtained by using the resistance values from hand to foot instead of from hand to hand.

(Choice D) A 100-times greater heat deposition results from comparing hand to foot under dry conditions and hand to hand under wet conditions instead of hand to hand for both dry and wet conditions.

Educational objective:
The heat energy deposited in the body is proportional to the electric energy dissipated. The heat energy is related to the ratio of the voltage squared and the resistance.

Time spent: 0 sec	**QID:** 403734
Subject: Physics	**System:** Electrostatics and Circuits

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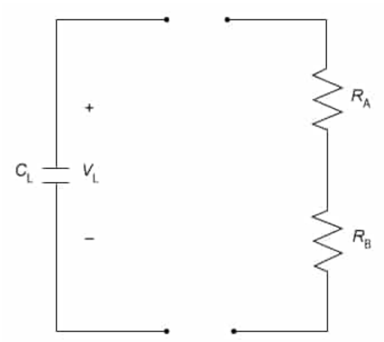

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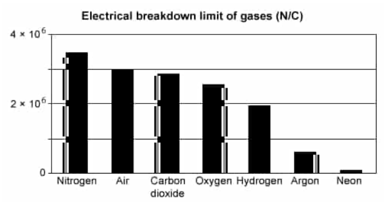


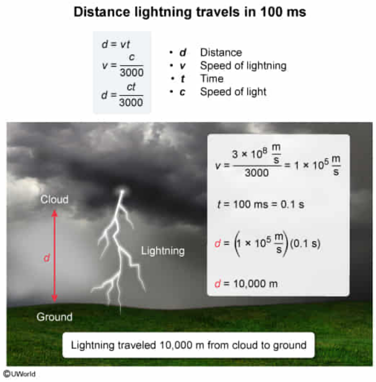
Figure 2 Breakdown limit of common gases (in N/C)

Lightning travels from a cloud to the ground in 100 ms. What is the distance the lightning traveled? (Note: The speed of light is $c = 3 \times 10^8$ m/s.)

- A. 10,000 m [39%]
- B. 3,000 m [47%]
- C. 1,000 m [7%]
- D. 300 m [5%]

Correct
39% answered correctly

Explanation:



The **distance** *d* traveled equals the product of **speed** *v* and **time** *t*:

$$d = \mathrm{vt}$$

In this question, lightning travels from a cloud to the ground in 100 ms. This value of *t* is converted from milliseconds (ms) to seconds (s) using the conversion 1 s = 1,000 ms:

$$t = (100\ \cancel{\mathrm{ms}})\left(\frac{1\ \mathrm{s}}{1{,}000\ \cancel{\mathrm{ms}}}\right) = 0.1\ \mathrm{s}$$

The passage states that lightning is 3,000 times slower than the speed of light *c*. Hence, *v* equals the ratio of *c* and 3,000:

$$v = \frac{3 \times 10^8\ \frac{\mathrm{m}}{\mathrm{s}}}{3{,}000} = \frac{1 \times 10^8\ \frac{\mathrm{m}}{\mathrm{s}}}{10^3} = 1 \times 10^5\ \frac{\mathrm{m}}{\mathrm{s}}$$

Therefore, d is equal to:

$$d = \mathrm{vt} = \left(1 \times 10^5\ \frac{\mathrm{m}}{\cancel{\mathrm{s}}}\right)(0.1\ \cancel{\mathrm{s}}) = 1 \times 10^4\ \mathrm{m} = 10{,}000\ \mathrm{m}$$

(Choice B) 3,000 m is obtained if the velocity of lightning is incorrectly calculated as 3×10^4 m/s.

(Choice C) 1,000 m is calculated if 0.01 s is used for the travel time of lightning.

(Choice D) 300 m results from using 3×10^3 m/s for the velocity of lightning.

Educational objective:
The distance traveled is equal to the product of the speed and the time.

Time spent: 0 sec
Subject: Physics

QID: 403735
System: Electrostatics and Circuits

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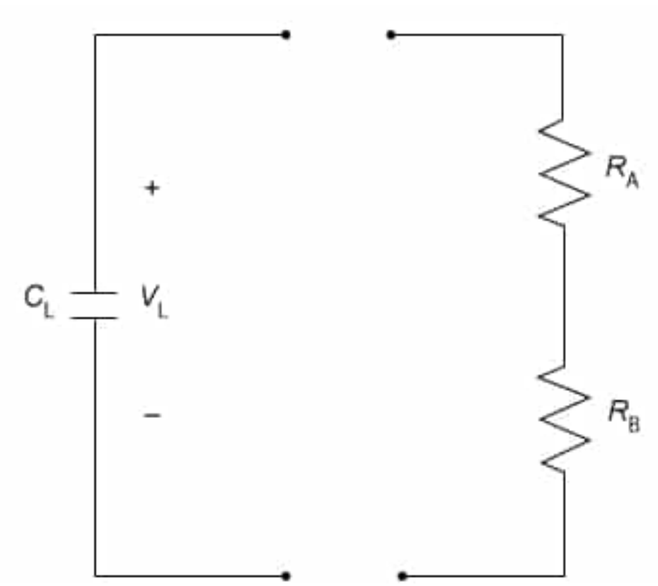

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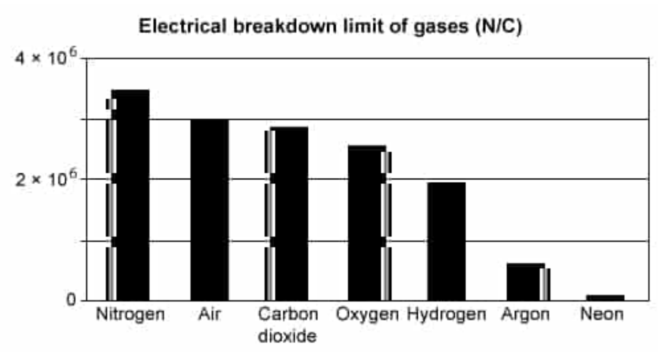


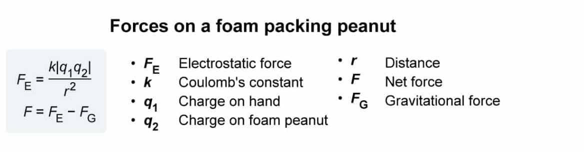
Figure 2 Breakdown limit of common gases (in N/C)

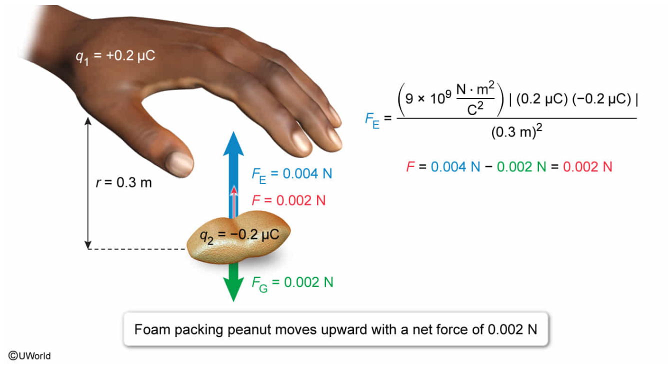
A foam packing peanut with a charge of −0.2 μC is held 30 cm below a person's hand that has a charge of +0.2 μC. Which of the following describes the motion of the packing peanut when it is released? (Note: Coulomb's constant is $k = 9 \times 10^9$ (N·m^2)/C^2.)

- A. Away from the hand with a net force of 0.02 N. [6%]
- B. Away from the hand with a net force of 0.002 N. [14%]
- C. Toward the hand with a net force of 0.02 N. [30%]
- D. Toward the hand with a net force of 0.002 N. [48%]

Correct
49% answered correctly

Explanation:





Coulomb's law states that the **electrostatic force** F_E between two objects with charges q_1 and q_2 can be calculated from q_1, q_2, Coulomb's constant k, and the **distance** r between the objects according to the equation:

$$F_E = \frac{k|q_1 q_2|}{r^2}$$

The charges repel with force F_E when q_1 and q_2 have the same sign, but the charges attract with the same force F_E when q_1 and q_2 have opposite signs.

In this question, a foam packing peanut is subjected to both F_E and the gravitational force F_G. The direction of F_G is downward, but the direction of F_E is upward because the hand is above the foam peanut and q_1 and q_2 have opposite signs. Hence, the net force F on the foam peanut equals the difference between F_E and F_G:

$$F = F_E - F_G$$

A positive F acts in the upward direction, and a negative F acts in the downward direction.

The passage states that F_G on a foam peanut equals:

$$F_G = 2\ \text{mN} = 0.002\ \text{N}$$

The electrostatic force F_E on the foam peanut can be calculated from Coulomb's law:

$$F_E = \frac{\left(9 \times 10^9\ \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}\right)|(0.2\ \mu\text{C})(-0.2\ \mu\text{C})|}{(30\ \text{cm})^2}$$

Converting all terms to standard units (meters and Coulombs) and simplifying gives:

$$F_{\mathrm{E}} = \frac{\left(\cancel{9} \times 10^{9}\ \frac{\mathrm{N}\ \cdot\ \cancel{\mathrm{m}^2}}{\cancel{\mathrm{C}^2}}\right)\left(2 \times 10^{-7}\ \cancel{\mathrm{C}}\right)\left(2 \times 10^{-7}\ \cancel{\mathrm{C}}\right)}{\cancel{9} \times 10^{-2}\ \cancel{\mathrm{m}^2}}$$

$$F_{\mathrm{E}} = 4 \times 10^{(9\,-\,7\,-\,7\,+\,2)}\ \mathrm{N} = 0.004\ \mathrm{N}$$

Accordingly, *F* equals:

$$F = 0.004\ \mathrm{N} - 0.002\ \mathrm{N} = 0.002\ \mathrm{N}$$

Therefore, the foam packing peanut will move upward toward the hand with a net force of 0.002 N.

(Choices A and B) The electrostatic force is greater than the gravitational force, so the foam peanut moves toward the hand, not away from the hand.

(Choice C) The foam peanut moves toward the hand, but the difference between the electrostatic and gravitational forces is 0.002 N, not 0.02 N.

Educational objective:
The electrostatic force between two charged objects is calculated from the equation $F_{\mathrm{E}} = \frac{k|q_1 q_2|}{r^2}$. If F_{E} is greater than the force of gravity, one of the objects could move toward the other instead of falling to the ground.

Time spent: 0 sec | **QID:** 403736
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